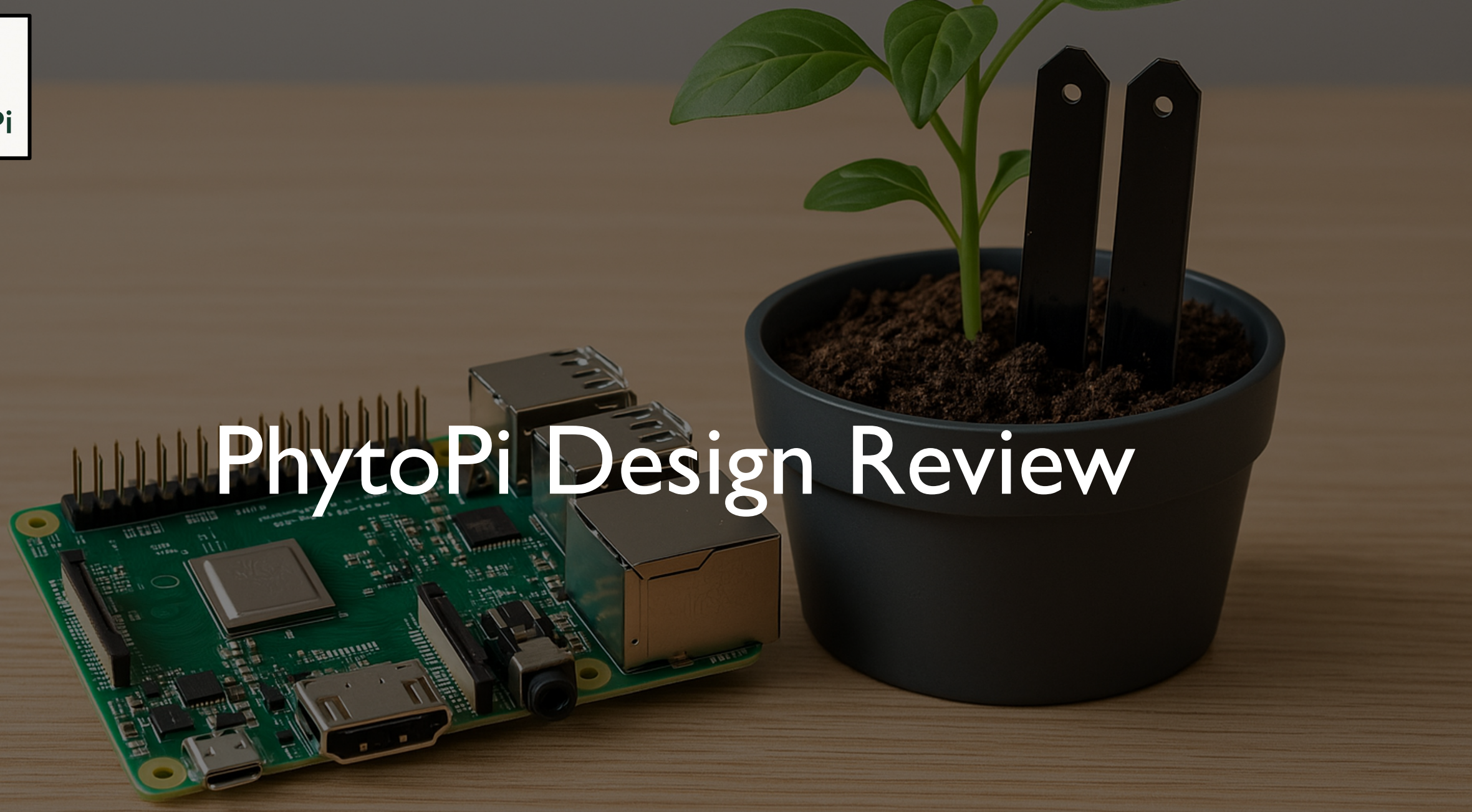
PhytoPi

PhytoPi Design Review

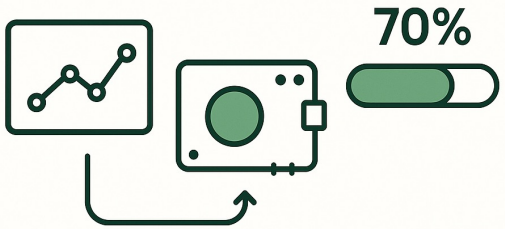


Nolan Tuttle, Daniel Grijalva

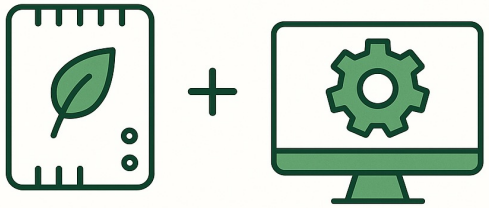
AGENDA

Project Summary

Current Progress

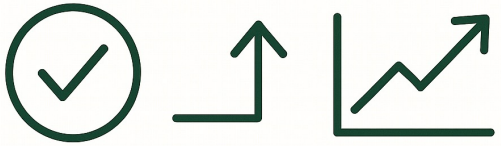


Initial Design Concepts



Functional Requirements

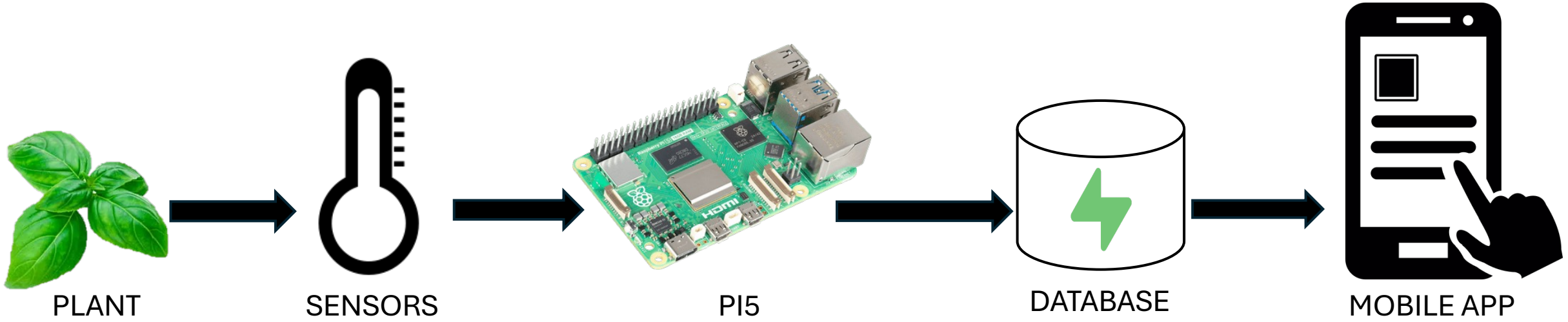
Next Steps



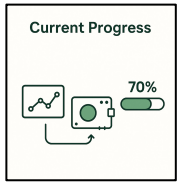
References & Questions

PROJECT SUMMARY

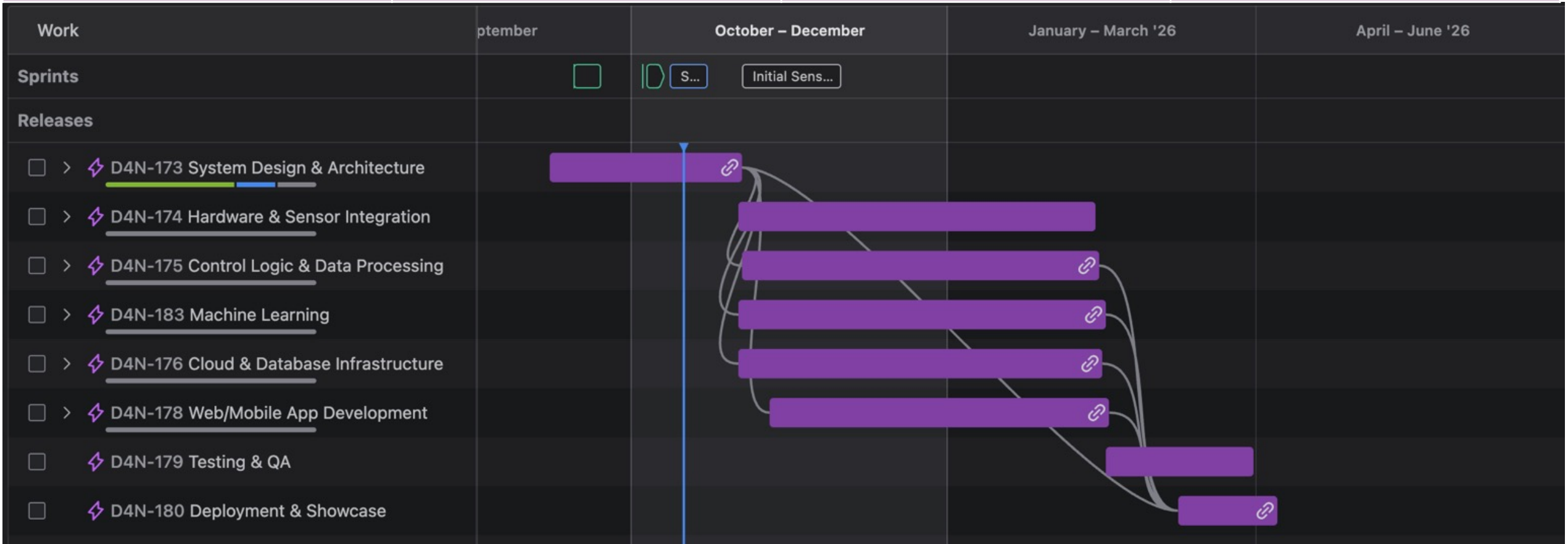
What is PhytoPi?	Goals	Core Features
A smart IoT ecosystem that monitors and automates plant growth using Raspberry Pi 5, Supabase, and Flutter.	• Hands-free plant care • Detailed analytics about plant health • Real-time plant health alerts and monitoring	• Soil moisture, humidity, temp, and light statistics • Responsive dashboards & alert notifications • AI plant analysis and automatic environment adjustment



Current Progress



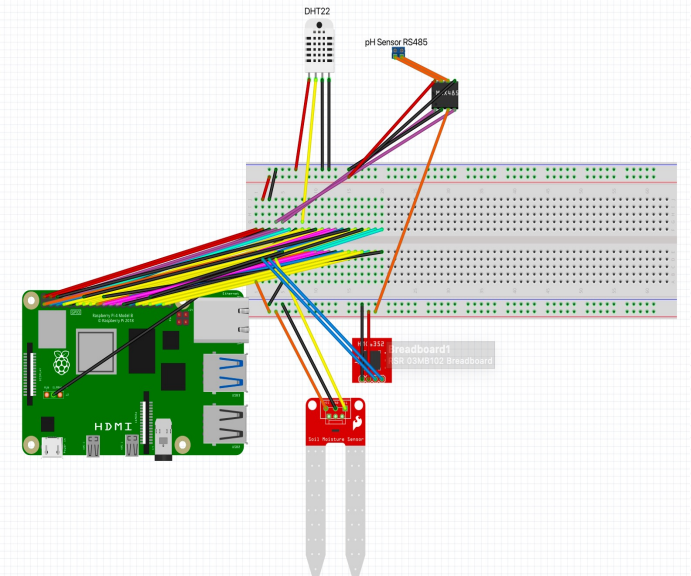
Data Layer	Business Logic Layer	UI Layer	Hardware Layer
Entity Relationship Diagram & Schemas Drafted	Planning Complete, Starting Initial Sensor Testing	Initial Wireframes Complete	Initial CAD & Electrical Design Complete

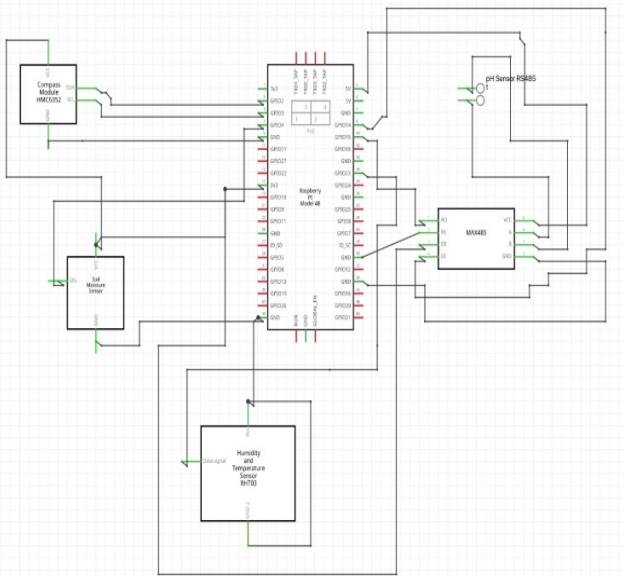


Initial Design Concept (Hardware)

Hardware Stack

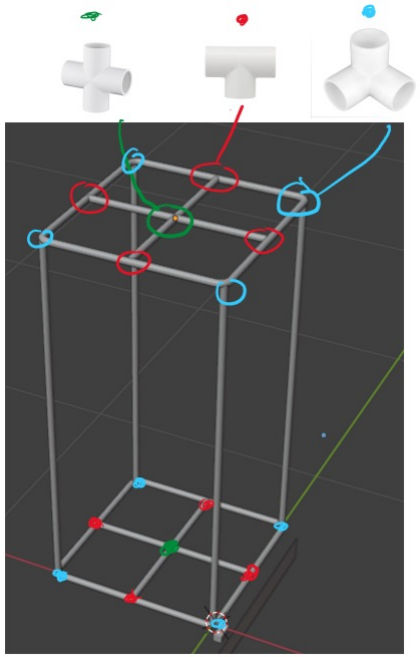
- Raspberry Pi 5
  - DHT22 (Temp/Humidity)
  - Capacitive Soil Humidity Sensor
  - Light Sensor
  - Raspberry Pi Camera Module



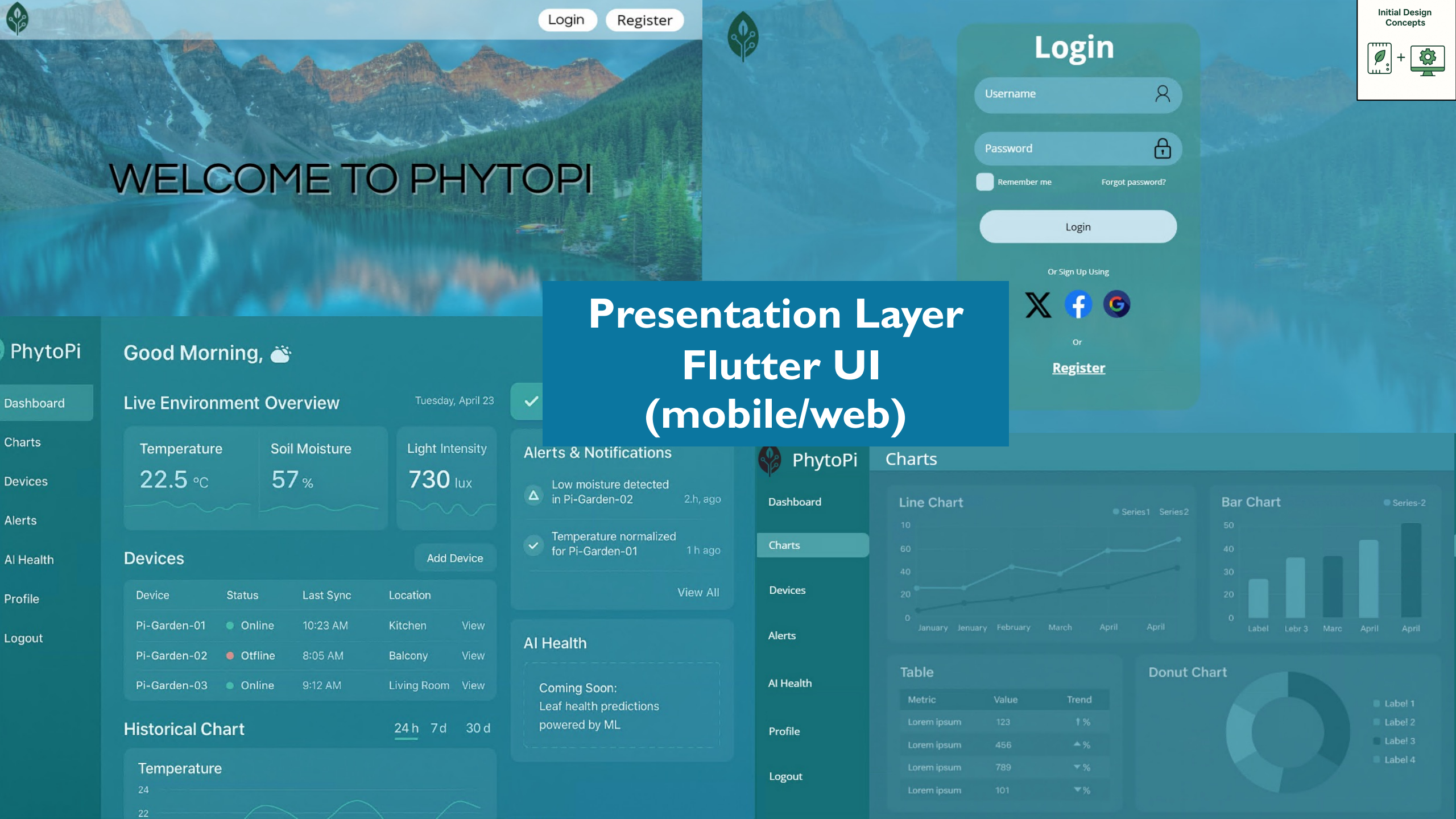


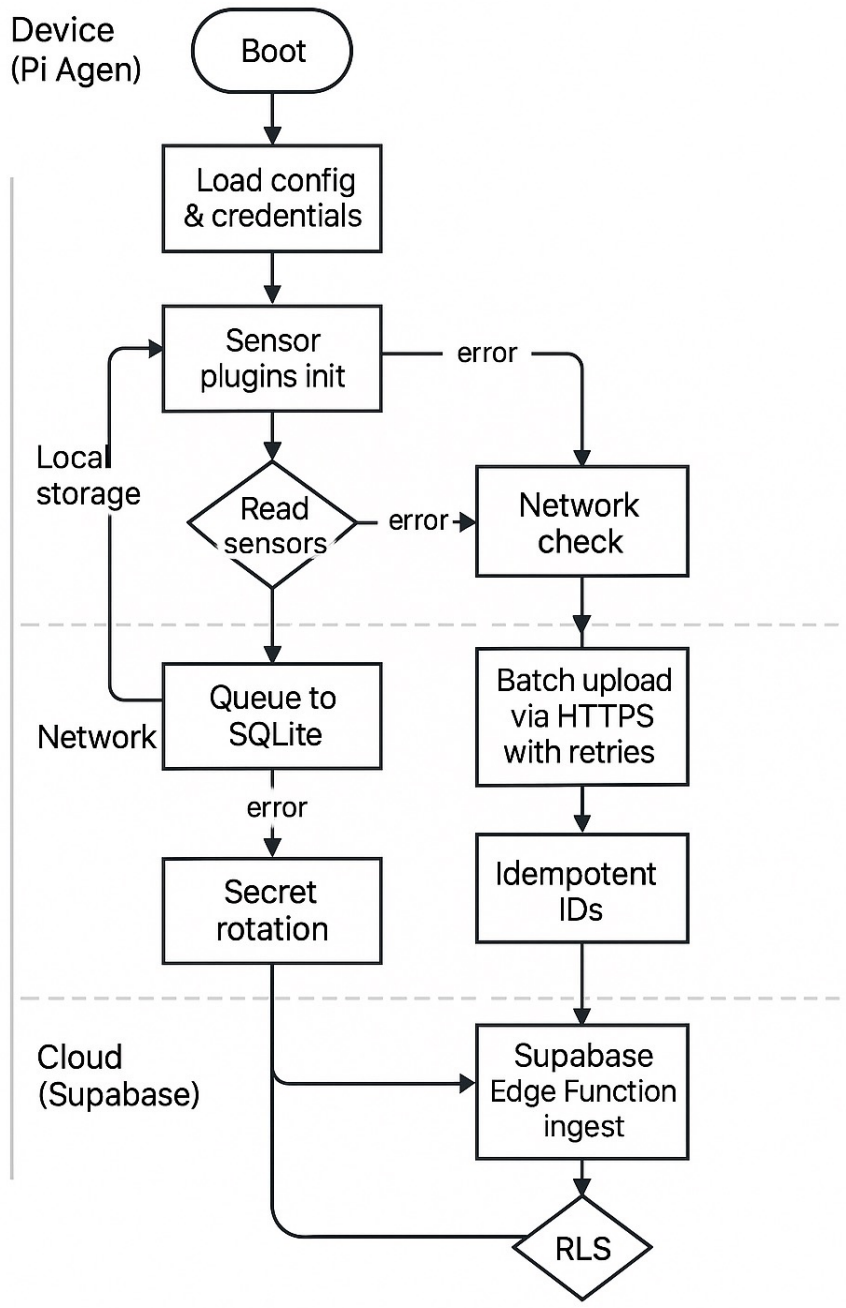
Mechanical Design

- PVC-based Frame
  - Waterproofed tent-like fabric enclosure
  - Sensors and power delivery integrated



Presentation Layer
Flutter UI
(mobile/web)

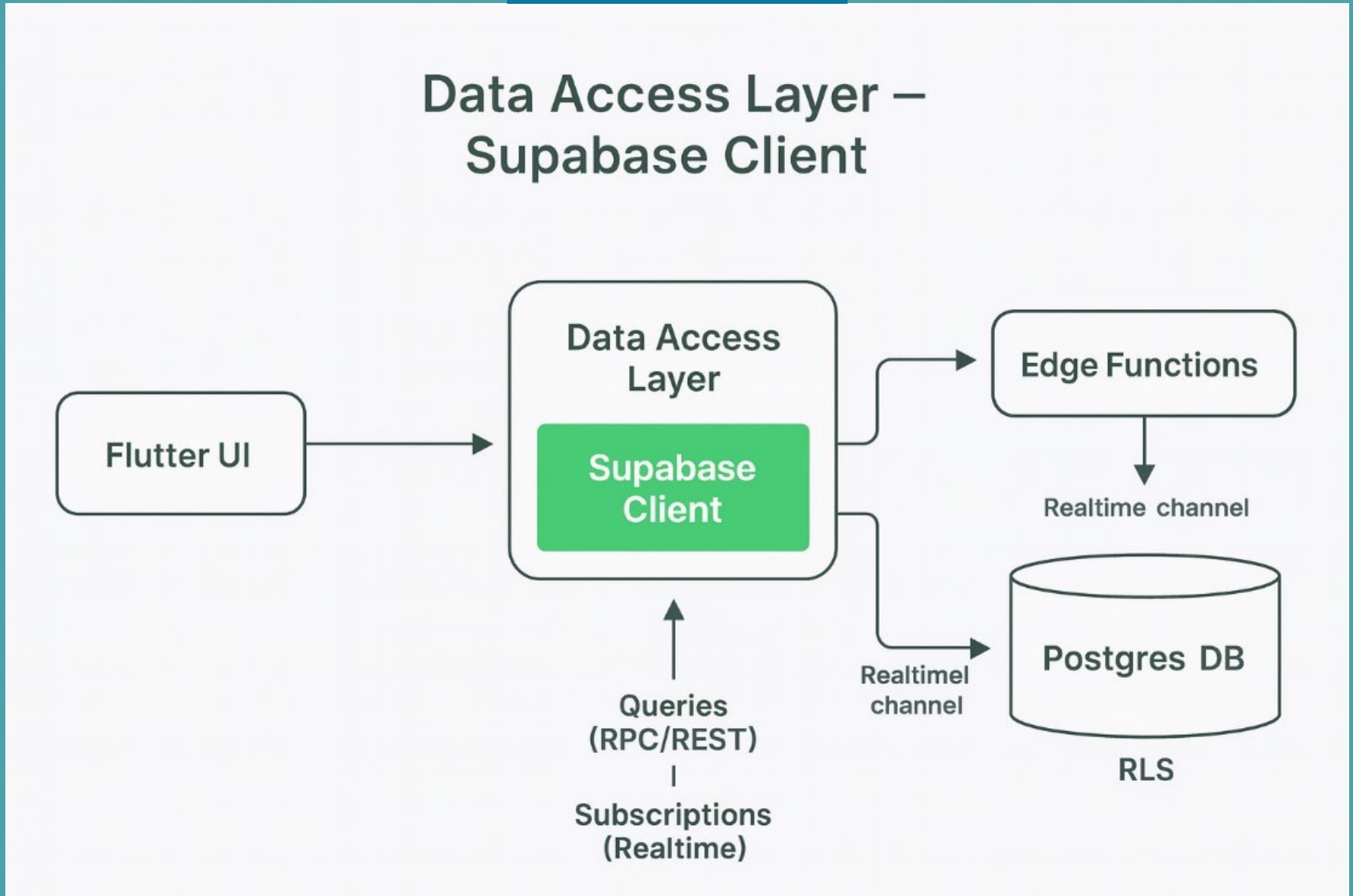




Business Logic Layer

Dart services for data processing & alerts

Data Access Layer
Supabase client (queries, subscriptions)



Data Layer
Supabase DB (PostgreSQL + Edge Functions)

Key Requirements & Standards

Functional Requirements

ID	Requirement	Relation to Customer Feedback	Engineering Standard
D4N-200	Monitor soil moisture, temp, humidity	Users want system automation for plant care	IEEE 2413-2019 (Architectural Framework)
D4N-208	Secure sensor-to-cloud communication	Privacy concern for IoT data	IEEE 29119 (Testing)
D4N-219	Mobile alerts for thresholds	Users want real-time updates on plant status	ISO 9126 (Usability)
D4N-221	Testing sensor data collection thresholds	Users want reliable performance and good availability	IEEE 2700-2017(Design Documentation)

Next Steps

Next Steps

- Begin Initial Sensor Testing Sprint
- Test Database with Sample Sensor Readings
- Finish System Design

References &
Questions

References

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- **PhytoPi Multiplatform App Roadmap (to MVP & Beyond).** (2025). Internal Documentation. Unpublished project document authored by N. Tuttle & D. Grijalva.

Questions?